

Concept Hierarchy Generation

- ❑ **Concept hierarchy** organizes concepts (i.e., attribute values) hierarchically and is usually associated with each dimension in a data warehouse
- ❑ Concept hierarchies facilitate drilling and rolling in data warehouses to view data in multiple granularity
- ❑ Concept hierarchy formation: Recursively reduce the data by collecting and replacing low level concepts (such as numeric values for *age*) by higher level concepts (such as *youth, adult, or senior*)
- ❑ Concept hierarchies can be explicitly specified by domain experts and/or data warehouse designers
- ❑ Concept hierarchy can be automatically formed for both numeric and nominal data—For numeric data, use discretization methods shown

Concept Hierarchy Generation for Nominal Data

- ❑ Specification of a partial/total ordering of attributes explicitly at the schema level by users or experts
 - ❑ $street < city < state < country$
- ❑ Specification of a hierarchy for a set of values by explicit data grouping
 - ❑ $\{Urbana, Champaign, Chicago\} < Illinois$
- ❑ Specification of only a partial set of attributes
 - ❑ E.g., only $street < city$, not others
- ❑ Automatic generation of hierarchies (or attribute levels) by the analysis of the number of distinct values
 - ❑ E.g., for a set of attributes: $\{street, city, state, country\}$